VELOOP REWARDS - Analytics Readiness Assessment

Data Analyst Internship, Task 01 | Work Package R

Authenticity notice: Nothing here is rated verified or implemented, because nothing has been confirmed with VELOOP yet. Levels describe the state of the work, not results. Counts taken from the supplied simulated dataset are labelled as such; no actual VELOOP statistic appears here.

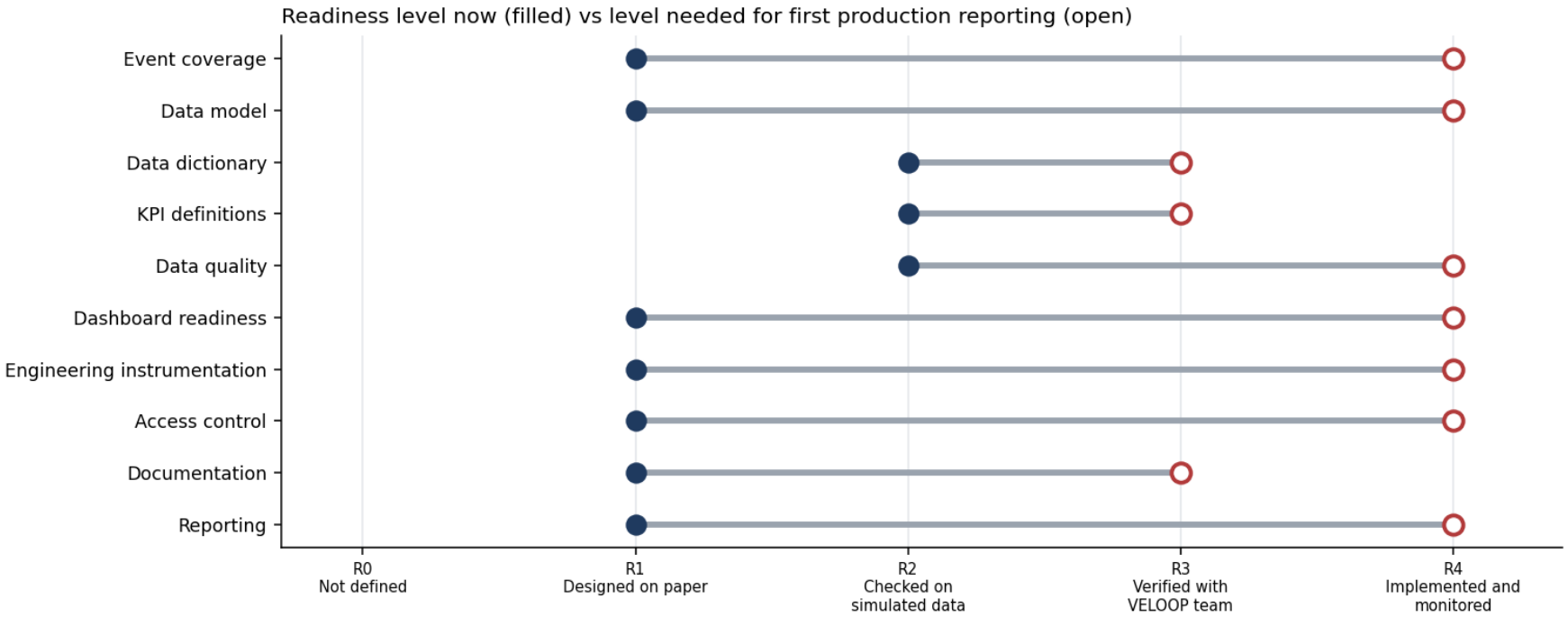
1. Summary

- **The documentation foundation is in place; production readiness is not.** 7 areas stand at level R1 (designed on paper) and 3 at R2 (checked against the supplied simulated dataset: data dictionary, KPI definitions and data quality). None has been verified with VELOOP (R3) or implemented (R4). The dataset covers the Users entity only, so R2 applies to Users.
- **31 gaps** across the 10 areas the brief lists: 3 Critical, 15 High, 10 Medium, 3 Low. Each has a priority reason, an owner, the assumptions it depends on, the evidence that would close it and a proposed backlog item.
- **The Critical gaps are RG-01 (events not confirmed to exist), RG-04 (stable user_id and transaction_id) and RG-19 (idempotency).** They block nearly everything else, so they come first.
- **Several gaps are decisions, not engineering work** (for example the definition of 'active' and the activation rule) and can be closed quickly by Product.

2. Readiness scale

Level	Meaning
R0	Not defined
R1	Designed on paper (this project)
R2	Designed and checked against the simulated dataset
R3	Verified with VELOOP engineering or product
R4	Implemented and monitored in production

3. Readiness scorecard



Levels describe the state of the work, not results. No VELOOP figures are shown.

Scorecard detail

Area	Needed Eventually	Currently Available / Verified	Evidence From This Project	Now	Needed	Priority	Priority Reason
Event coverage	Every event in the plan that supports a live feature is emitted, validated and de-duplicated; at least the Critical and High events	No event is confirmed to exist in VELOOP	Event Tracking Plan (42 events, 9 Critical); Coverage Matrix (42 rows); Developer Handoff (25 event requirements)	R1	R4	Critical	Without events, no production KPI can be computed
Data model	The 10 entities stored with stable keys and enforced relationships	Logical model proposed; not compared with VELOOP's real tables	Logical Data Model (16 foreign keys, 20 relationships, ER diagram)	R1	R4	Critical	Without a stable user_id and transaction_id, joins and wallet deduplication cannot be guaranteed
Data dictionary	Every field defined with confirmed allowed values, units and sensitivity	128 fields defined; 0 confirmed; the Users entity was compared with the supplied dataset; other entities have no data	Data Dictionary (128 fields, 99 unique names, 6 Restricted); Simulated Data Profiling (schema alignment of the 27 dataset columns)	R2	R3	High	Allowed values, units and methods are unconfirmed, so validation cannot be finalised
KPI definitions	Agreed, reproducible definitions with owners and no unsettled terms	54 KPIs defined; 12 To Be Verified; tested against the supplied file: 2 calculable, 5 partly, 47 not; 'active' and 'activation' undecided	KPI Catalogue (54 KPIs); Simulated Data Profiling (KPI test of every KPI)	R2	R3	High	Core KPIs could change meaning once 'active' and 'activation' are decided
Data quality	Rules running at the earliest sensible layer, with quarantine, owners and rules proven on real data	39 rules designed; 18 tested on the Users data (7 passed, 8 failed); 21 could not be tested	Data Quality Control Framework (39 rules, 11 Critical); Data Quality Audit (25 issues logged)	R2	R4	High	Only the Users data could be tested; rules for rewards, wallet, withdrawals, referrals and events remain unproven
Dashboard readiness	Dashboards fed by the analytics layer, with a chosen tool and confirmed KPIs	56 visuals specified; no tool, no data	Dashboard Blueprint (7 dashboards, 56 visuals, wireframes)	R1	R4	Medium	Specified, but blocked by events; building earlier would add no value
Engineering instrumentation	Collection route, idempotent writes, UTC timestamps, provider ids and a data store	Specification written; no engineering response recorded	Developer Handoff (25 events, 120 payload rows, 12 general requirements, 12 pre-conditions); Pipeline Proposal (13 components)	R1	R4	Critical	Idempotency and the collection route are undecided, so duplicates and gaps are possible
Access control	Roles, owners, grants and logging in place; Restricted fields protected	Access model proposed for 6 roles and 10 zones; nothing implemented	Privacy and Governance Proposal (6 Restricted fields, owners, workflow)	R1	R4	High	Restricted fields need owners and grants before data flows
Documentation	Reviewed, versioned and agreed with Product and Engineering	22 documents at baseline versions; not yet reviewed by the VELOOP team	Assumption Register (50 unknowns), Change Management (log, versions, impact matrix)	R1	R3	Medium	Statuses stay Proposed until the VELOOP team confirms them
Reporting	Regular business reporting from governed production data	No production reporting exists or is verified	KPI Catalogue and Dashboard Blueprint define what reports would contain	R1	R4	High	No figure can be relied on until the layers below exist

4. Gaps by area and priority

Area	Critical	High	Medium	Low	Total
Event coverage	1	1	1	0	3
Data model	1	1	1	0	3
Data dictionary	0	1	0	1	2
KPI definitions	0	3	1	0	4
Data quality	0	2	2	0	4
Dashboard readiness	0	1	2	0	3
Engineering instrumentation	1	3	1	0	5
Access control	0	1	1	1	3
Documentation	0	0	1	1	2
Reporting	0	2	0	0	2
Total	3	15	10	3	31

5. Gap register

ID	Area	Gap	Priority	Priority Reason	Owner	Assumptions	Needed By	Closing Evidence	Proposed Backlog Item
RG-01	Event coverage	None of the 42 proposed events is confirmed to exist in VELOOP, including the 9 Critical ones	Critical	Without registered, task, reward, wallet and withdrawal events, none of the core KPIs can be computed from production data	Engineering	A001, A021	Pre-launch	Engineering lists emitted events with a sample record for each	Confirm which events exist and who emits them
RG-02	Event coverage	It is not confirmed which events come from the frontend and which from the backend	High	Outcomes emitted by the frontend would count results the system never confirmed	Engineering	A021	Pre-launch	Emitter recorded for every Critical and High event	Confirm emitter for each event
RG-03	Event coverage	17 Medium and Low events are not specified to engineering level	Medium	Not needed for the first release, but needed for full coverage of features and dashboards	Data Analyst	A044	Post-launch	Handoff requirements written for the remaining events	Specify Medium and Low events
RG-04	Data model	A stable user_id and a unique transaction_id are not confirmed	Critical	Without a stable user_id, cross-event analysis breaks; without a transaction_id, reliable wallet deduplication cannot be guaranteed	Engineering	A003, A005	Pre-launch	Engineering confirms immutable user_id and unique, idempotent transaction_id	Stable user_id and transaction_id
RG-05	Data model	Withdrawal state history may not be stored with a timestamp per state	High	Without per-state times, only the latest status is visible and processing time cannot be measured	Engineering	A006, A007	Pre-launch	Sample withdrawal shows requested, processing and final timestamps	Store withdrawal state history
RG-06	Data model	The logical model has not been compared with VELOOP's real tables	Medium	Entities and keys follow the brief, not the product, so some may need to change	Data Analyst	A011, A031	Pre-launch	Model compared with the real schema and differences logged	Compare model with real schema
RG-07	Data dictionary	All 128 fields are Proposed or To Be Verified; none is Confirmed	High	Allowed values, units and methods are placeholders, so validation rules cannot be finalised	Engineering / Product	A002, A006, A022	Pre-launch	Fields confirmed with allowed value lists in the reference table	Confirm fields and approved lists
RG-08	Data dictionary	Example values are illustrative, so real value formats are unknown	Low	Formats matter for validation but do not block design	Engineering	A022	Post-launch	Real examples (sanitised) added	Add real example formats
RG-09	KPI definitions	Which events make a user 'active', and the activity window, are not decided	High	DAU, WAU, MAU, returning and inactive users all depend on it	Product	A024	Stakeholder decision	Decision recorded in the change log	Decide 'active' definition
RG-10	KPI definitions	The activation definition for users and referrals is not agreed	High	Referral activation and conversion would be reported on an unstable rule	Product	A010	Stakeholder decision	One activation rule agreed with a version	Agree activation definition
RG-11	KPI definitions	Timezone and week and month boundaries are not agreed	Medium	Daily metrics shift between days if the timezone changes	Engineering / Product	A012	Pre-launch	Timezone and boundaries written into the KPI Catalogue	Agree timezone and period boundaries

ID	Area	Gap	Priority	Priority Reason	Owner	Assumptions	Needed By	Closing Evidence	Proposed Backlog Item
RG-12	KPI definitions	Reward units and any conversion between them are not defined	High	Amounts in different units cannot be added or compared	Product / Finance	A002	Pre-launch	Unit list and conversion rule confirmed	Define reward units
RG-13	Data quality	The quality rules have been run only on the Users dataset; rules that need other datasets could not be tested	High	Detection logic for reward, wallet, withdrawal, referral and event rules is unproven, and most KPIs could not be tested because their data was not supplied	Mentor / Product team; Data Analyst	A046	Pre-launch	Remaining datasets supplied and the audit and KPI test re-run	Obtain remaining datasets and re-run profiling, audit and KPI tests
RG-14	Data quality	Freshness, volume, clock tolerance and late-arrival limits are not agreed	Medium	Monitoring rules need limits to alert on	Engineering / Data Analyst	A035	Pre-launch	Limits agreed and written into the framework	Agree monitoring limits
RG-15	Data quality	No owner is assigned to triage the quarantine table	Medium	Quarantined records stay unresolved and KPIs stay incomplete	Data Analyst / Product	A034	Pre-launch	Named triage owner and cadence	Assign quarantine triage owner
RG-16	Dashboard readiness	Dashboards cannot be built from production data until events and tables exist	High	The blueprint is ready but there is no data to show; building early would show empty or misleading charts	Data Analyst	A021	Post-launch	First dashboards built on confirmed Critical events	Build first dashboards after Critical events
RG-17	Dashboard readiness	The BI tool is not chosen	Medium	Aggregated-only access and refresh depend on the tool	Product / Data Analyst	A032	Pre-launch	Tool chosen against the blueprint and access model	Choose BI option
RG-18	Dashboard readiness	12 KPIs are To Be Verified and need a caveat or to be hidden until confirmed	Medium	Showing unverified KPIs without a label would overstate certainty	Data Analyst	A010, A013, A029, A030	Before the related dashboard	Each KPI confirmed or labelled on the dashboard	Label or confirm To Be Verified KPIs
RG-19	Engineering instrumentation	No idempotency design is agreed for event and wallet writes	Critical	Retries could create duplicate events and duplicate wallet credits	Engineering	A005, A023	Pre-launch	Idempotency approach documented and tested with a repeated request	Agree and test idempotency
RG-20	Engineering instrumentation	The collection route and analytics data store are not chosen	High	Events have nowhere to land and the pipeline cannot start	Engineering	A011, A033	Pre-launch	Route and store documented; a test event reaches the raw zone	Choose collection route and store
RG-21	Engineering instrumentation	Provider reference ids and callbacks are not confirmed	High	Third-party earning cannot be reconciled and completions may be understated	Engineering	A009, A042	Pre-launch	Provider-validated attempts carry a reference id	Store provider reference ids
RG-22	Engineering instrumentation	Timestamp format and timezone in the backend are not confirmed	High	A wrong day or a missing timezone changes daily metrics and streaks	Engineering	A012	Pre-launch	Sample records show UTC ISO 8601	Standardise UTC timestamps
RG-23	Engineering instrumentation	A pending reward state and the grain of task attempts are not confirmed	Medium	Pending Rewards and Task Completion Rate depend on them	Engineering	A020, A031	Pre-launch	Engineering confirms both	Confirm pending state and attempt storage
RG-24	Access control	Roles, data owners and access grants exist only on paper	High	Without them, Restricted fields (6) could be over-exposed and approvals have no owner	Product / Engineering	A037, A039	Pre-launch	Named owners and an access log	Name data owners and set up access
RG-25	Access control	Retention, deletion and consent approaches need Legal review	Medium	Raw data is append-only, so deletion requests need a designed process	Legal / Product	A016, A036	Pre-launch	Written Legal guidance and a retention period per zone	Get Legal guidance on retention and deletion
RG-26	Access control	Small-group suppression size and access review cadence are not set	Low	Needed for safe aggregated reporting but not blocking	Product	A038, A039	Post-launch	Values agreed	Set suppression size and review cadence
RG-27	Documentation	The documents have not been reviewed with Product and Engineering	Medium	Statuses stay Proposed until the VELOOP team confirms them	Data Analyst	A018	Pre-launch	Review notes and statuses updated	Review documents with the team
RG-28	Documentation	Change approval roles and notice periods are not agreed	Low	The change process works without them at first	Product	A040, A041	Post-launch	Roles and periods recorded	Agree change approvers and notice
RG-29	Reporting	No production reporting exists or is verified	High	Management cannot rely on any figure until events, tables and quality checks are in place	Data Analyst / Product	A021, A043	Post-launch	First report produced from governed production data	Produce first governed report
RG-30	Reporting	Simulated and production data separation is designed but not implemented	High	Mixing simulated and production data would produce false results (rule DQ039)	Data Analyst / Engineering	A011	Pre-launch	Separate zones in place; load of simulated data into production blocked	Implement simulated and production separation

ID	Area	Gap	Priority	Priority Reason	Owner	Assumptions	Needed By	Closing Evidence	Proposed Backlog Item
RG-31	Data quality	Stored derived fields, counters and consent and status fields in the supplied Users data contradict each other in places	High	Metrics built on these fields would be wrong; several of the issues exist in the 'clean' sheet as well as the audit sheet	Engineering / Product	A047, A049, A050	Pre-launch	Derived values computed in the analytics layer; counters and consent rules enforced and tested	Fix derived fields, counters and consent consistency

6. Where to start

- Ask Engineering to confirm the three Critical gaps; most close with one written answer each.
- Ask Product for the decisions (RG-09, RG-10, RG-12) because they are quick and unblock several KPIs.
- Provide the remaining datasets so profiling, the KPI test and the audit can be re-run (RG-13, A046).

7. Open questions

- Which Critical gaps can Engineering answer first (RG-01, RG-04, RG-19)?
- When can the remaining datasets be provided, so the data quality gap (RG-13) can close (A046)?
- Who will review the document set for the VELOOP team (RG-27)?
- Which Stakeholder decisions can be taken now ('active' definition, activation rule)?

Next: the Final Implementation Backlog, built from these gaps.